

Biochip Course

Experimental Lab

Instructions

Marco Carminati

a.y. 2022/2023

March 29th 2023



Tasks (mandatory):

- Design electrodes and PCB (simulation of impedance)
- Perform measurements in liquid and data analysis

Tasks (optional):

- Circuit simulations (e.g. LTSpice)
- Design of fluidics

Expected learning outcomes:

- Develop hands-on experience
- Develop autonomous project organization skills
- Experience teamwork
- Develop presentation skills



Experimental Lab Activity

Goal: design circuit to measure properties of liquid/solid interface

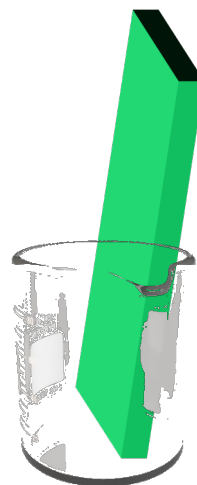
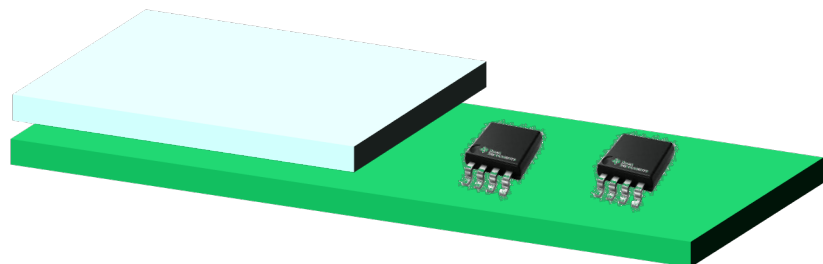
Electrodes

Front-End
Electronics



PCB

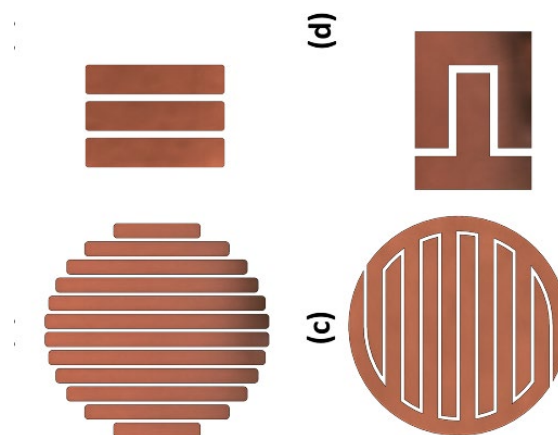
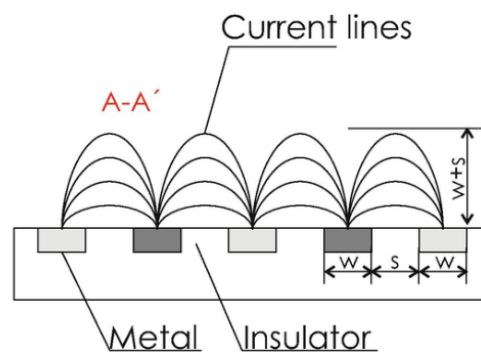
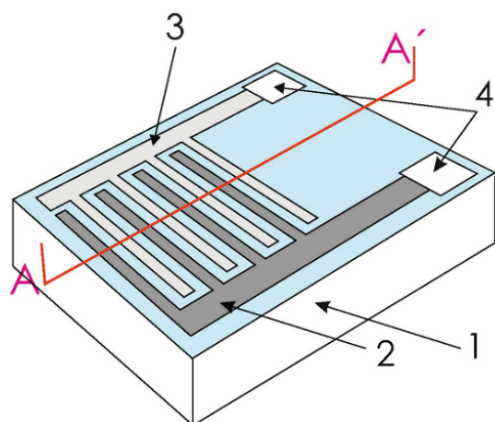
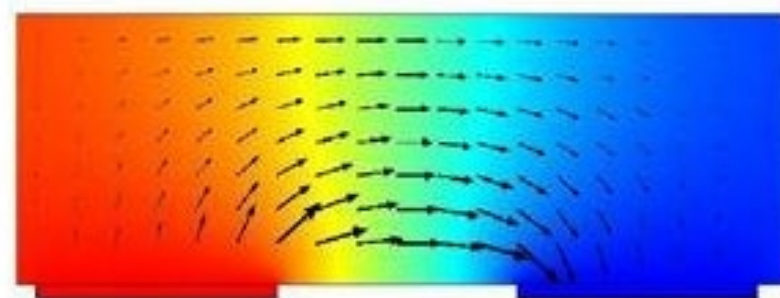
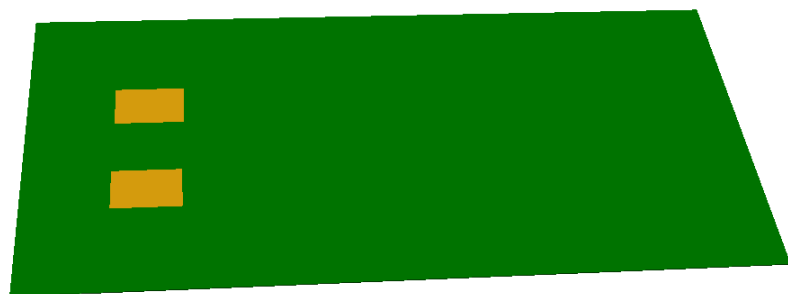
Microfluidics





1. Coplanar Electrodes

Electrodes will be gold-plated and coplanar: you can choose the size (area, distance) and shape





2. Estimation of Coplanar Impedance

Use conformal mapping to estimate R_{SOL} :



Available online at www.sciencedirect.com

SCIENCE @ DIRECT®

Sensors and Actuators A 112 (2004) 291–301

SENSORS
AND
ACTUATORS
A
PHYSICAL

www.elsevier.com/locate/sna

Analytical evaluation of the interdigital electrodes capacitance for a multi-layered structure

Rui Igreja*, C.J. Dias

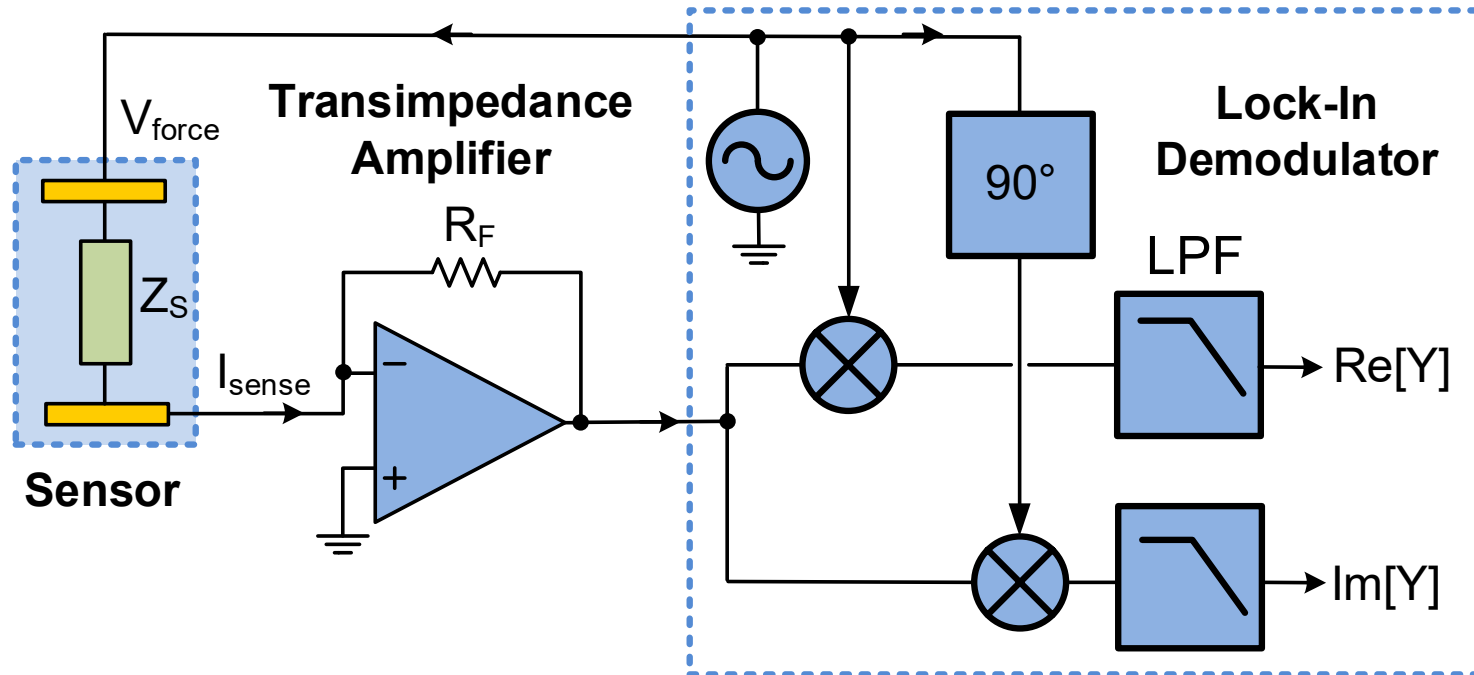
*Departamento de Ciência dos Materiais/CENIMAT, Faculdade de Ciências e Tecnologia, Universidade Nova de Lisboa,
Campus da Caparica, 2829-516 Caparica, Portugal*

Received 14 May 2003; received in revised form 14 January 2004; accepted 30 January 2004

- Starting point: tap water resistivity: in Milano $700\mu\text{S}/\text{cm}$
- Add ions (salt)
- Test also de-ionized water



3. Front-End Transimpedance Amplifier



Constraints:

- Dual voltage supply: $\pm 5\text{V}$
- Instrument (Moku:Go) input: $\pm 5\text{V}$
- OpAmp AD8022 (SOIC-8 SMT package)
- Spectroscopy f from 1kHz to 1MHz (10MHz), R_{SOL} visible



USB Instrument: Moku:Go Liquid Instruments

Analog Inputs/Outputs
Two 12 bit, 125 MSa/s

Input Bandwidth
30 MHz

Digital I/O
16 channel @ 62.5 MSa/s

Output Bandwidth
20 MHz

Programmable Power Supplies
2 or 4 channel option

11 Powerful Instruments

- Arbitrary Waveform Generator
- Data Logger
- Digital Filter Box
- Frequency Response Analyzer
- FIR Filter Builder
- Logic Analyzer / Pattern Generator
- Oscilloscope / Voltmeter
- PID Controller
- Spectrum Analyzer
- Waveform Generator
- Lock-in Amplifier* (add-on option)

Programmable Power Supplies

2 channel option

- +5 V to -5 V @ 150 mA
- 0 to 16 V @ 150 mA

4 channel option

- +5 V to -5 V @ 150 mA
- 0 to 16 V @ 150 mA
- Dual 0.6 to 5 V @ 1 A

Specifications

Analog Inputs

- Two 12 bit, 125 MSa/s input channels
- 30 MHz analog bandwidth
- AC or DC coupling with 1 M Ω impedance
- Input range up to ± 25 V

Analog Outputs

- Two 12 bit, 125 MSa/s output channels
- 20 MHz analog bandwidth
- ± 5 V maximum output range

Digital I/O

- 16-channel DIO at 62.5 MSa/s
- Support 3.3 V (5 V tolerant) logic level

Programming Environment

- API support for Python, MATLAB, and LabVIEW
- Windows or MacOS

Models

M0, \$599

- 2 analog inputs, 2 analog outputs and 16 DIO
- USB-C, Wi-Fi, software, and APIs

M1, \$699

- All features from M0
- Two-channel programmable power supply

M2, \$799

- All features from M0
- Ethernet
- Four-channel programmable power supply

Options & Accessories

- *Lock-in Amplifier, \$200
- All models include relevant accessories: 2 oscilloscope probes, DIO cabling, power adapter, USB-C and Ethernet cable, and power supply cables
- 6 standard colors, or custom color upgrade





2. Dual OpAmp



Dual High Speed,
Low Noise Op Amp

AD8022

FEATURES

Low power amplifiers provide low noise and low distortion,
ideal for xDSL modem receiver

Wide supply range: +5 V, ± 2.5 V to ± 12 V voltage supply

Low power consumption: 4.0 mA/Amp

Voltage feedback

Ease of Use

Lower total noise (insignificant input current noise
contribution compared to current feedback amps)

Low noise and distortion

2.5 nV/ $\sqrt{\text{Hz}}$ voltage noise @ 100 kHz

1.2 pA/ $\sqrt{\text{Hz}}$ current noise

MTPR < -66 dBc (G = +7)

SFDR 110 dB @ 200 kHz

High speed

130 MHz bandwidth (-3 dB), G = +1

Settling time to 0.1%, 68 ns

50 V/ μs slew rate

High output swing: ± 10.1 V on ± 12 V supply

Low offset voltage, 1.5 mV typical

FUNCTIONAL BLOCK DIAGRAM

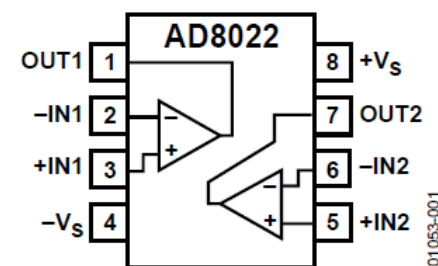


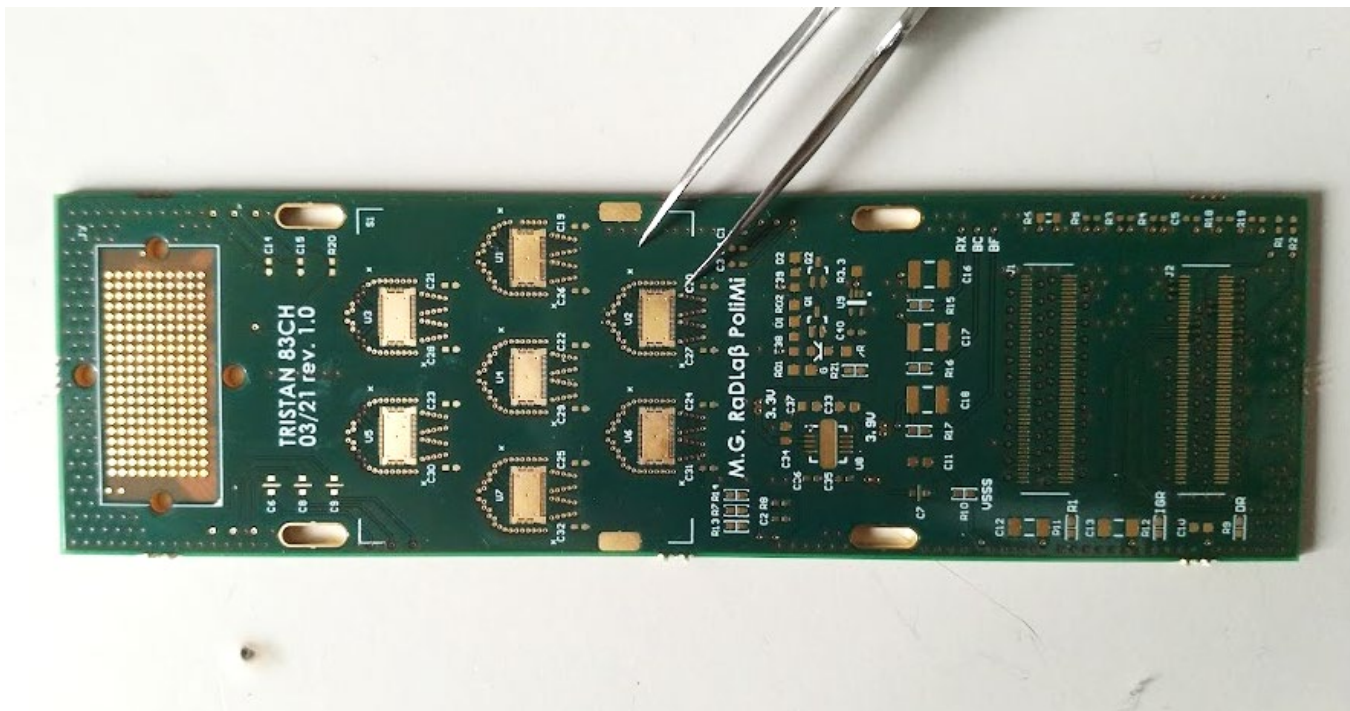
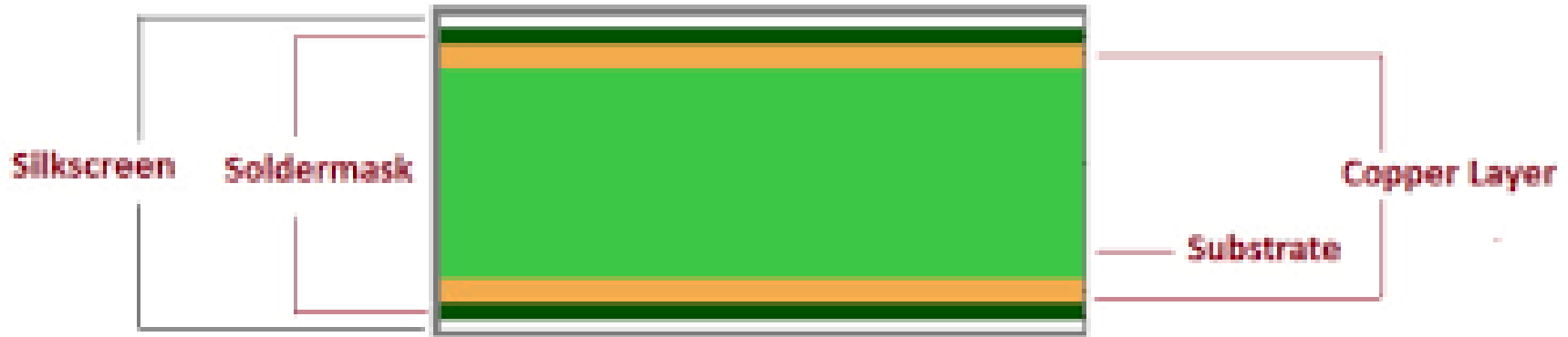
Figure 1.

Two opamps in one
package

Add decoupling
capacitors at V_s pins



2-Layer PCB



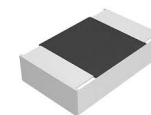
Gold
surface
finish



Board Size



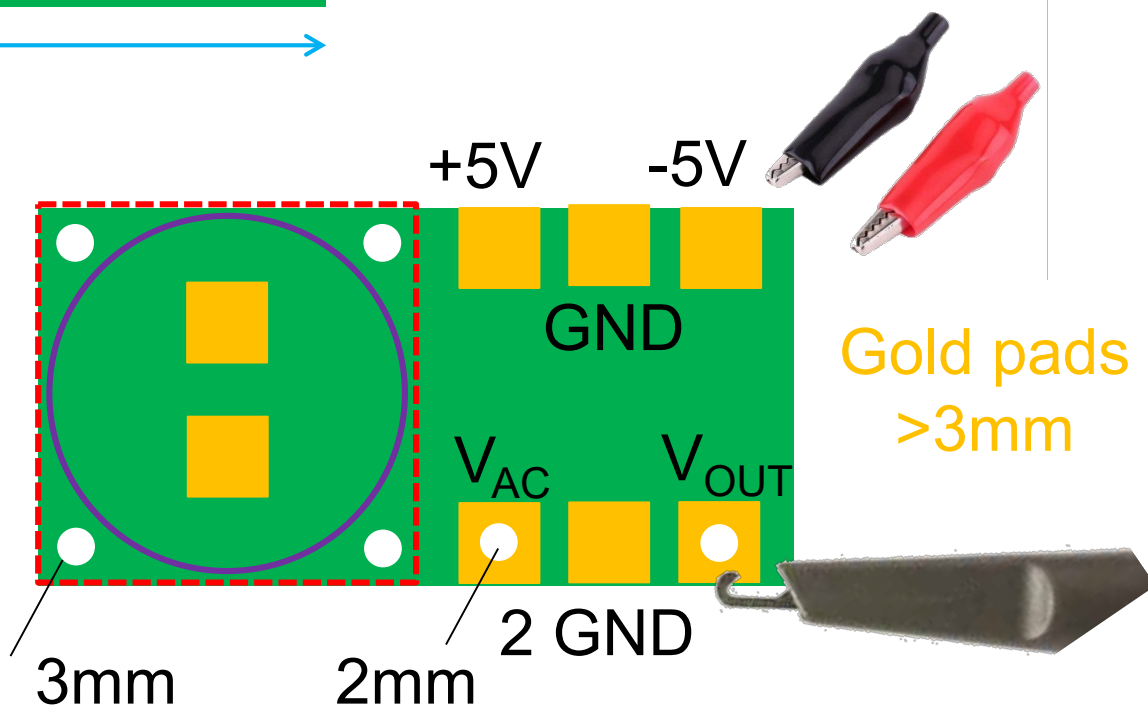
0805 SMD package
for passive
components



Pads for connections

Region to be coupled
with fluidics

O-ring 26mm ID,
28mm OD





PCB Design with KiCAD

Free software <https://www.kicad.org/download/>

The screenshot shows the KiCad website's download page. At the top is the KiCad logo and a navigation menu with links: BLOG, DISCOVER, COMMUNITY, HELP, CONTRIBUTE, SPONSORS, LIBRARIES, DOWNLOAD, and ABOUT. Below the menu is a banner image showing various PCB designs. A breadcrumb trail reads 'Home / Download'. The main heading is 'Download', followed by the instruction 'Select your operating system'. There are three large buttons for selection: 'Windows' with the Windows logo, 'macOS' with the Apple logo, and 'Linux/Other' with the Tux penguin logo.

Tutorial:

<https://www.youtube.com/watch?v=aVUqaB0IMh4>



Project Already Created

Ki lab_2023 — KiCad 7.0

File Modifica Visualizza Strumenti Preferenze Aiuto

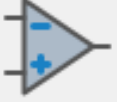


File progetto

- lab_2023.kicad_pro
 - lab_2023-backups
 - lab_2023_lib.pretty
 - lab_2023.kicad_pcb
 - lab_2023.kicad_sch



Editor degli schemi
Modifica lo schema elettrico del progetto



Editor simboli
Modifica le librerie di simboli globali e/o del progetto



Editor C.S.
Modifica il C.S. del progetto



Editor impronte
Modifica le librerie di impronte globali e/o del progetto



Visualizzatore Gerber



Project Already Created

lab_2023 — Editor C.S.

File Modifica Visualizza Inserisci Sbrogli Ispeziona Strumenti Preferenze Aiuto

Pista: usa larghezza netclass Via: usa dimensioni netclass Griglia: 2.5400 mm (0.1000 in) Zoom 8.00

F.Cu (PgUp)

Aspetto

Strati Oggetti

- F.Cu
- B.Cu
- F.Silkscreen
- B.Silkscreen
- F.Mask
- B.Mask
- Edge.Cuts
- Margin
- F.Courtyard
- B.Courtyard
- F.Fab
- B.Fab

Opzioni di visualizzazione

Preimpostazioni (Ctrl+Tab):

Viste (Shift+Tab):

Filtro selezione

- ☒ Tutti gli elementi
- ☒ Impronte
- ☒ Piste
- ☒ Piazzole
- ☒ Zone
- ☒ Dimensioni
- ☒ Elem
- ☒ Testc
- ☒ Via
- ☒ Grafi
- ☒ Aree
- ☒ Altri

Piazzole 19 Via 0 Segmenti di pista 0 Piste 9 Non sbrogliato 10

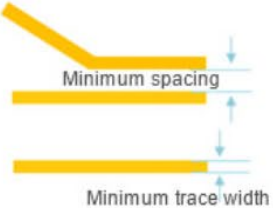
Z 7.06 X 182.8800 Y 83.8200 dx 182.8800 dy 83.8200 dist 201.1738 grid X 2.5400 Y 2.5400 mm



PCB Design Rules:

Minimum trace width and spacing



	Min. Trace width	Min. Spacing	Patterns
1-2 Layers	5mil (0.127mm)	5mil (0.127mm)	
4-6 Layers	3.5mil (0.09mm)	3.5mil (0.09mm)	
2oz Copper weight	8mil (0.2mm)	8mil (0.2mm)	

<https://jlcpcb.com/capabilities/Capabilities>



Fluidic Design (Optional)



Form 3 by FormLabs

25 μm resolution (x, y and z)

(take some margins)

Transparent resin

Suggested software:
Fusion360 by Autodesk



submit .stl file

Mandatory:

- Choose electrodes size and geometry
- Estimate impedance (by conformal mapping)
- Design TIA schematic (size R_F) and PCB layout
- Present and discuss your design
- Perform measurements in lab (changing ions concentration)

Optional:

- Spice simulations of the circuit (noise... [LTSpice](#))
- Propose **variants** (geometry, fluidics, 2 opamps...)
- Design of fluidics to be coupled to PCB
- Fitting of data for analysis



Deadlines

2 SEMESTRE

marzo 2023	aprile 2023	maggio 2023
1 mer	1 sab	1 lun
2 gio	2 dom	2 mar
3 ven	3 lun	3 mer Discussion
4 sab	4 mar	4 gio Upload
5 dom	5 mer	5 ven
6 lun	6 gio	6 sab
7 mar	7 ven	7 dom
8 mer MI	8 sab	8 lun
9 gio POLI	9 dom Pasqua	9 mar
10 ven	10 lun Lunedì dell'Angelo	10 mer
11 sab	11 mar	11 gio
12 dom	12 mer	12 ven
13 lun	13 gio	13 sab
14 mar	14 ven	14 dom
15 mer	15 sab	15 lun
16 gio	16 dom	16 mar
17 ven	17 lun	17 mer
18 sab	18 mar	18 gio
19 dom	19 mer	19 ven
20 lun	20 gio	20 sab
21 mar	21 ven	21 dom
22 mer	22 sab	22 lun
23 gio	23 dom	23 mar
24 ven	24 lun	24 mer
25 sab	25 mar	25 gio
26 dom	26 mer	26 ven
27 lun	27 gio	27 sab
28 mar	28 ven	28 dom
29 mer Start	29 sab	29 lun Lab 1
30 gio	30 dom	30 mar
31 ven		31 mer Lab 2

- May 3rd: class discussion
5-min presentation per team + 5-min discussion
Max. 5 slides (possible to anticipate to April 26th)
- May 4th: projects submission

**MAY THE
FORCE
BE WITH
YOU**

- **May 29 & 31: lab activity**



What to Address in the Presentation

Mandatory:

- Electrodes geometry
- Expected value of ionic resistance and spectrum
- Choice of R_F
- Ideas for test protocol to measure R_{ion}

Optional:

- Spice simulations of TIA
- Noise analysis
- Fluidics
- Other ideas



9 Teams & Submission

Teams:

1. Check teams members
2. Choose a name for each team
3. Create a group with teams members and start interacting

[BIOCHIP Lab Teams 2023.xlsx](#)

Submission Checklist:

1. **KiCAD file with layout** (no gerber)
2. **Value of R_F** and additional components if required
3. Optional: stl file for fluidics (or Fusion project)



Building 23, via Golgi, 42 (in front of admin offices)

